



SpaceX is ready

SpaceX announced that it is already ready to launch its next rocket just four months after its last tragic accident. <http://dgit.in/SpceXjan>



Gym vs Tsunami

Architects and engineers in Westport, Washington designed a gym that can withstand a tsunami. <http://dgit.in/GymTsunami>

Workshop

Speedtest. Do note, Twitch has restrictions to a video bitrate of 3,500 kbps (unless you're a partner). Facebook Live has many restrictions on encoding settings such as no support for hardware encoding, a maximum audio bitrate of 128, maximum resolution of 720p in 30 fps, and maximum video bitrate of 4,000 kbps.

Choosing whether you want to use software or hardware encoding depends on your rig. If you have a strong CPU like an Intel i7, then you can pick the x264 encoder with audio bitrate at 160 and Encoder Preset on fast. You'll need to check whether your CPU is capable of handling the different Encoder Presets where ultrafast doesn't stress the CPU whereas slower in intensive. You should definitely opt for x264 encoding if you don't have a discrete GPU.

But if you do have a decent or powerful GPU, you could enable hardware encoding and the corresponding encoder to your GPU (AMD or NVIDIA) will appear. Here, you can again experiment with the encoder presets from High Quality to Low-latency High Performance. In our testing, we found the hardware encoder to be better at streaming with the GTX 1080 even though we had the i7-3960X.

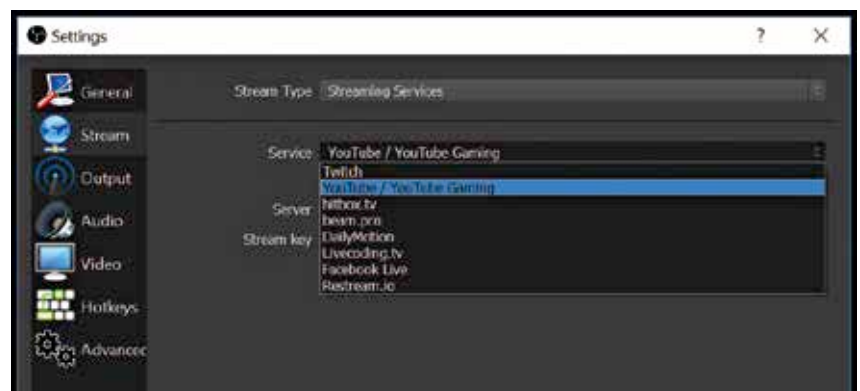
Audio and video settings

There isn't much in the audio and video settings since everything will be set to default. You can choose between the sample rates and reassign your desktop audio and mic sources. In audio, Studio sees an update over Classic to support multiple sources for desktop audio and mics. Click on the dropdown menu for Desktop Audio Device and Mic/Auxiliary Device, and select your devices rather than letting the default setting stay for better control over your stream.

The video settings essentially ask you to set the resolution and fps that you want to output on your stream. It could be 1920x1080 at 60fps or 1280x720 at 60fps, depending on your internet bandwidth and PC specs. Choose the Downscale Filter as Lanczos only if you have a powerful rig.

Previews, dry runs and going live

Before you go live to your audience, it's important to test whether scene transitions are functioning properly. You can right-click on each scene and go to Fullscreen Projector that will give you a preview on how that scene will look like in the stream. You should do this for every scene, just to be sure. The best way to check whether all your hotkeys work is to record a session. When all the scenes and sources are final, hit the Start Recording button and it will save a local copy of the entire session.



Different streaming services support different parameters

You can refer to the recorded video to discover flaws and then move on to a dry run online.

Do note that even the first dry run on your YouTube channel will send a notification to your subscribers (if you have any) that you've gone live. Hence, it would be advisable to use a secondary undisclosed YouTube channel to experiment.

After you've entered all the settings in YouTube as mentioned above, hit Start Streaming in OBS and the top bar in the YouTube Live dashboard should turn green or yellow or red. The different colours depict the quality of the stream where green is the best and red is the worst. A preview should appear in the window below where you can see the scenes from OBS. Start the game you want to stream and switch to that particular scene if you're using Game Capture. Since the game will run in full-screen, the best way to check whether your live stream is functional is to watch

it live from a mobile device, preferably on a different connection (cellular) or you could ask a friend to watch. You can try switching between the scenes and testing all the hotkeys, and also observe whether the stream suffers frame drops or freezes. If you notice these problems, then you'll have to go back to tweaking the encoder settings and more tests.

If you have a strong internet bandwidth that can handle both the streaming and watching simultaneously, then you can have the YouTube Live dashboard running on another monitor if you have one. This will

enable you to watch and reply to your chat messages and keep an eye on the stream's health. It is also essential to check your audio levels for each audio input source. Sometimes, your in-game audio could be overpowering your mic input and you will have to reduce the volume for your desktop or in-game audio source. You'll also need to check for dropped frames in your stream at the bottom and there are numerous reasons to blame for it. You can check out this comprehensive post that tries to solve this problem <http://dgit.in/OBSDroppedFrames>.

Once your live stream is lag-free, you're ready to go live and face the world to display your high level of skills or the lack of it. On a different note, we do stream the lack of our skills in CS: GO almost every day on SKOAR!'s official Facebook page. Make sure you head over and like the page so that you're get a notification whenever we go live.